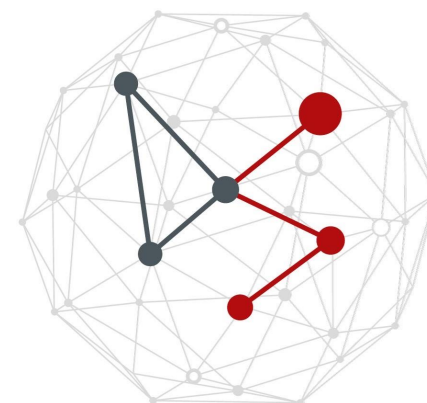


CNN AS FEATURE EXTRACTORS: THE CASE OF INERTIAL SIGNALS

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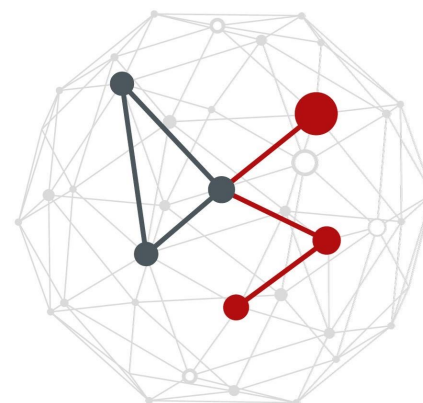


Outline

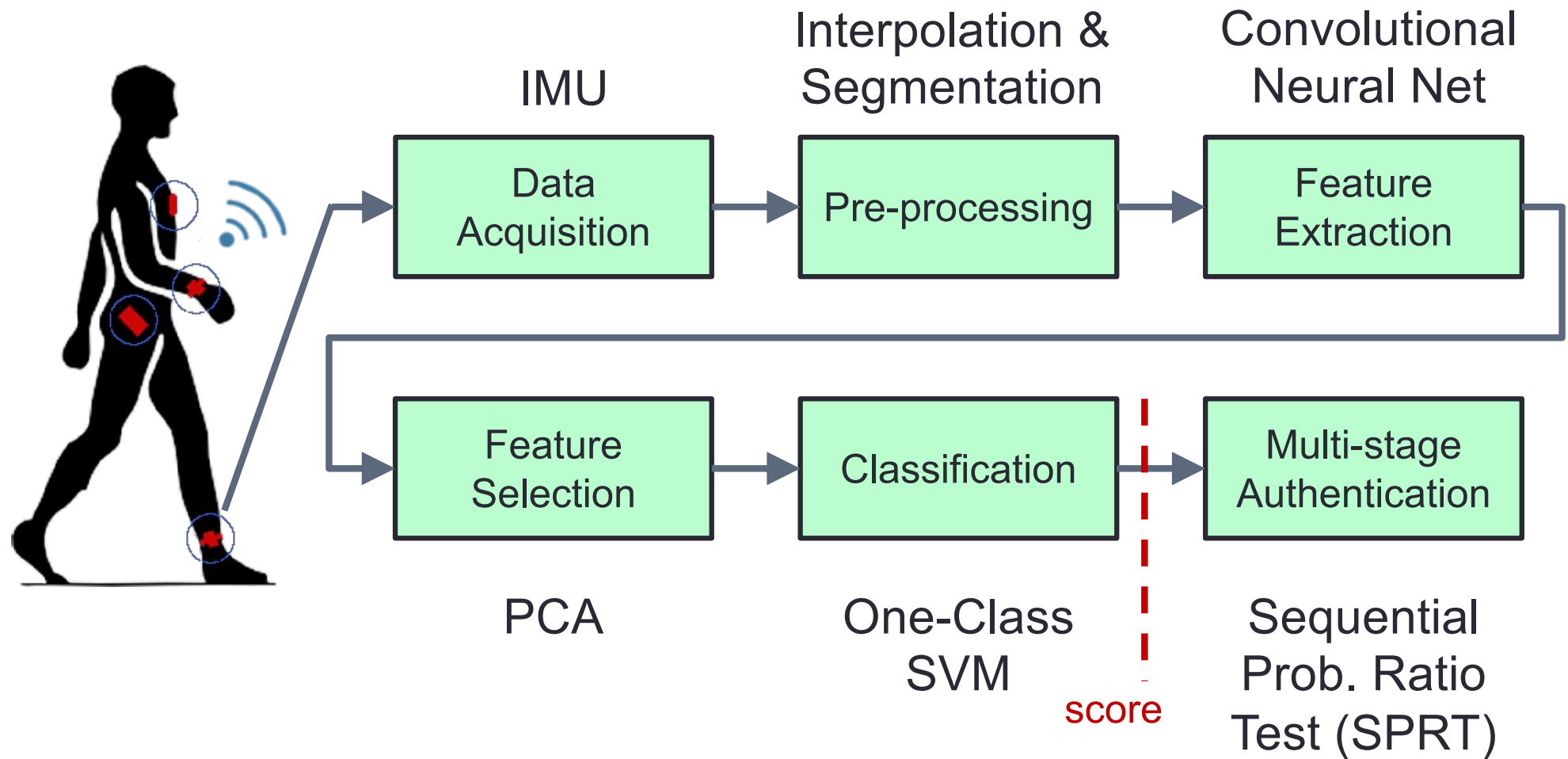
- Analysis of inertial signals
 - System model
 - Data acquisition & preprocessing
 - Coordinate transformation
 - Template based segmentation
 - CNN architecture
 - One-class SVM
 - Sequential decision making
 - Experimental results



ANALYSIS OF INERTIAL SIGNALS



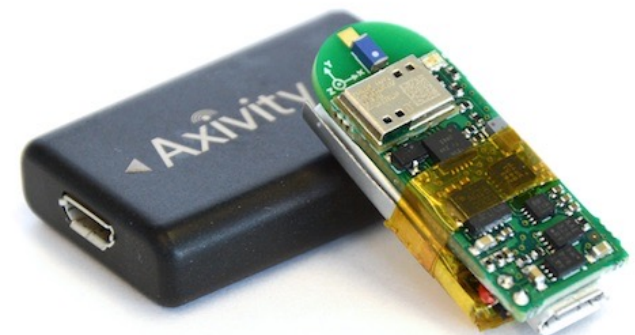
System Model



Data Acquisition (1/2)

Inertial Measurement Unit (IMU)

- **Axivity WAX9**
 - **Accelerometer $\pm 2 / 4 / 8$ g** (14 bit resolution)
 - **Gyroscope $\pm 250 / 500 / 2000$ dps** (16 bit resolution)
 - Temperature 0 - 65 °C (0.1 °C resolution)
 - Pressure 30-110 kPa (1Pa resolution)
 - Max. sampling frequency: **400 Hz**
 - Bluetooth LE radio
 - <https://axivity.com/downloads/wax9>
- **Worn on the ankle of the user**



Data Acquisition (2/2)

Smartphone

- Measures the 9 axes
- Sampling frequency
 - Fluctuates (non-realtime OS)
 - Some samples may be lost
 - Re-interpolation needed
- Carried in the front pocket



Pre-processing

Interpolation & resampling

- Cope with missing points & variable sample rate

Coordinate system transformation

- Onto rotation invariant system

Template extraction

- Manually done from acc & gyro magnitudes

Segment extraction

- Correlation measure between signal & template

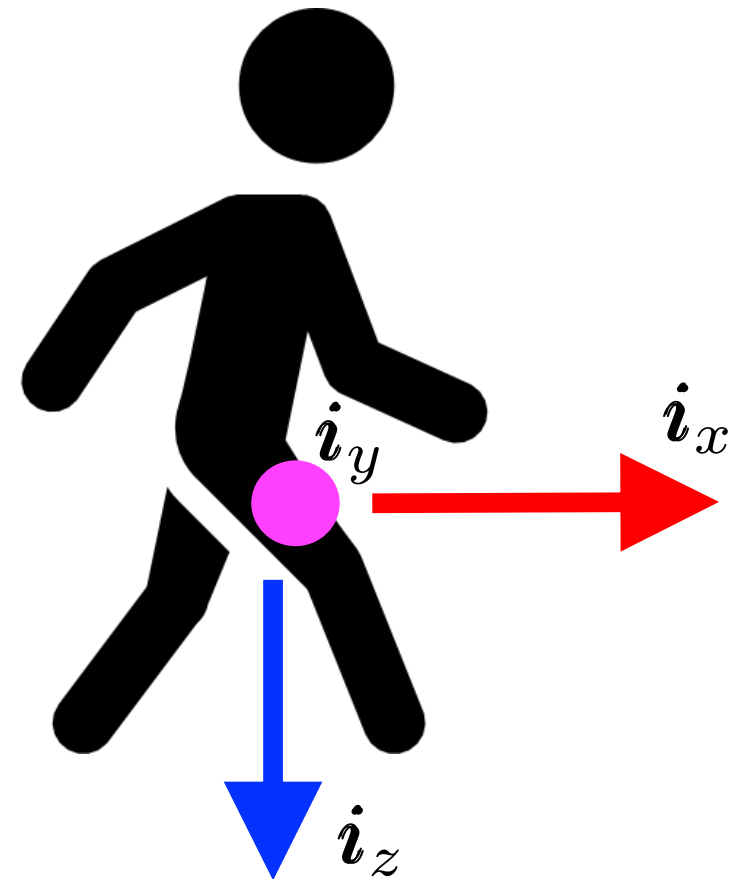
$$\text{corr_dist} = 2 - \text{corr}(\text{acc}) - \text{corr}(\text{gyro})$$

Transformation of coordinate system

- Only needed for smartphone system
- Accelerometer and gyroscope
 - Measured in the coordinate system of the smartphone
 - This depends on phone orientation in the pocket
 - Not good
- What we need
 - To measure a trajectory that is independent of the phone orientation
 - Orientation invariant coordinate system
- Solution
 - Move the x, y, z measurements (9 axes)
 - from phone system to orientation invariant coordinate system

Orientation invariant coordinate system

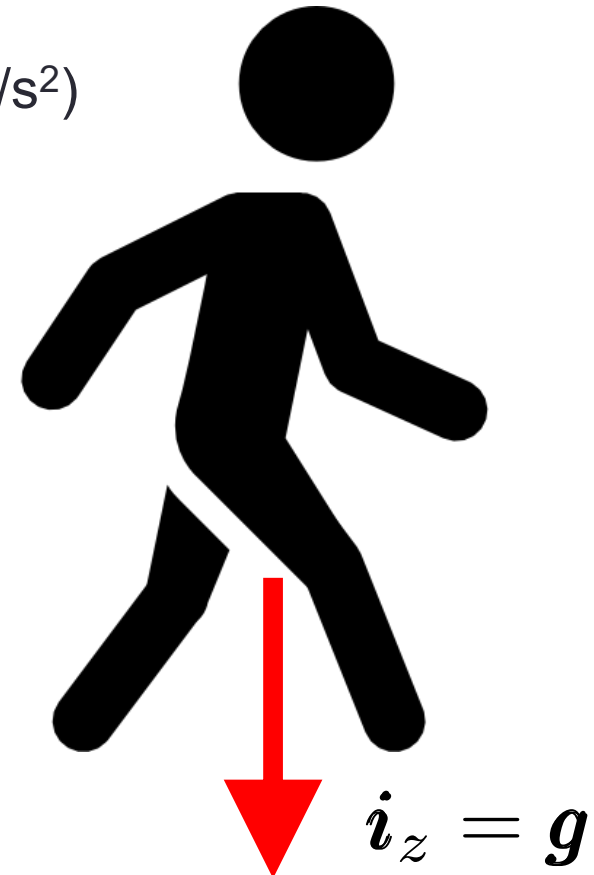
- 3 orthogonal versors are to be found
 - **One pointing forward**
 - Aligned with direction of motion
 - **One pointing down**
 - Aligned with gravity (and user's torso)
 - **One tracking lateral movement**
 - Orthogonal to the other two versors



Orientation invariant coordinate system

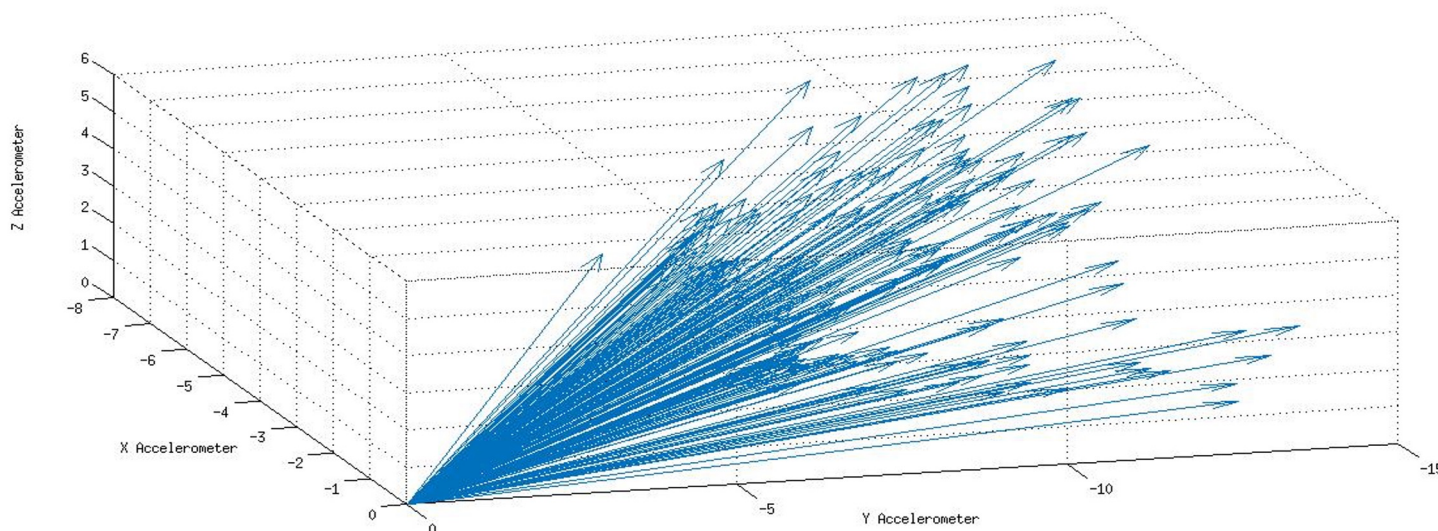
- Solution

- **Gravity!** $\mathbf{g} = (g_x, g_y, g_z)$
- Is a constant acceleration vector
- We even know its average value (9.81 m/s²)



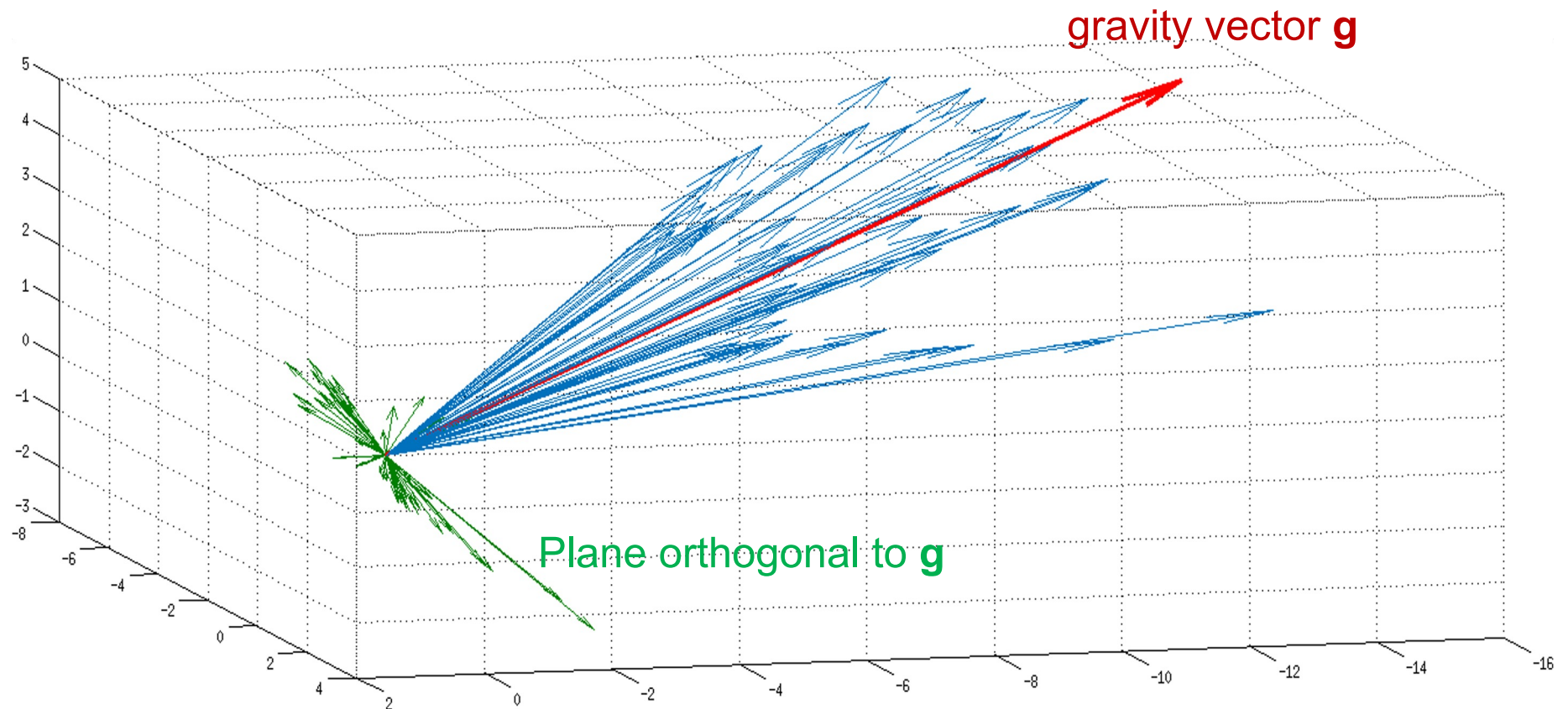
Orientation invariant coordinate system

- Obtaining the new coordinate system
 - Measure accelerometer signal (a_x , a_y , a_z), in the native system
 - Low pass filter (averaging)
 - this extracts constant gravity vector (constant, 0 Hz component)
 - This will be aligned with the first versor (pointing down)
 - Below graph
 - 3D acceleration signal



Low pass filtering and projection

- Low pass filtering to find $\mathbf{g}$
- Projection onto orthogonal plane to $\mathbf{g}$



Plane orthogonal to $\mathbf{g}$

- How to identify the forward direction of motion?

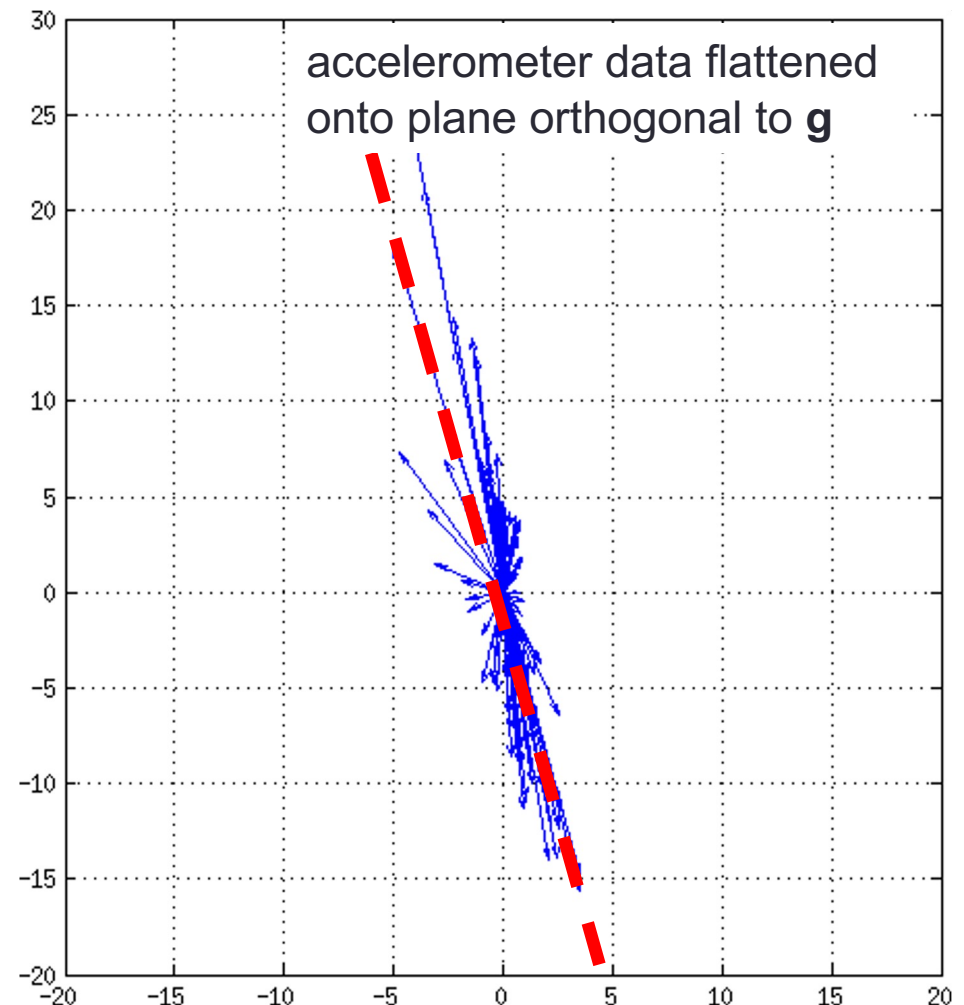
- Corresponds to dashed line

- Variance of data cloud

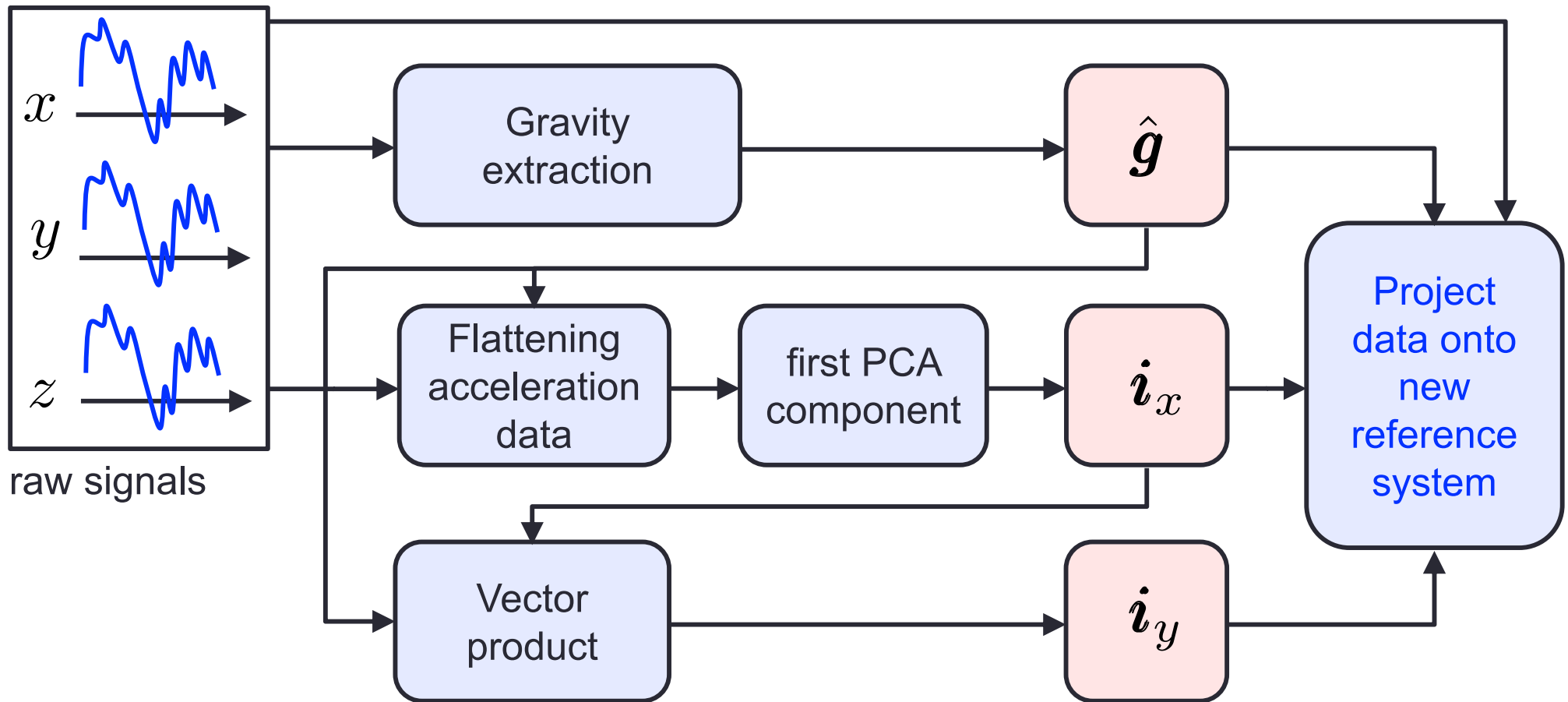
- Is max along direction of motion
- And once the alignment is found (dashed red line in right plot), the forward movement has largest projected variance wrt backward

maximum variance $\rightarrow$ PCA!!!

- First PCA component
- Has strongest projected variance

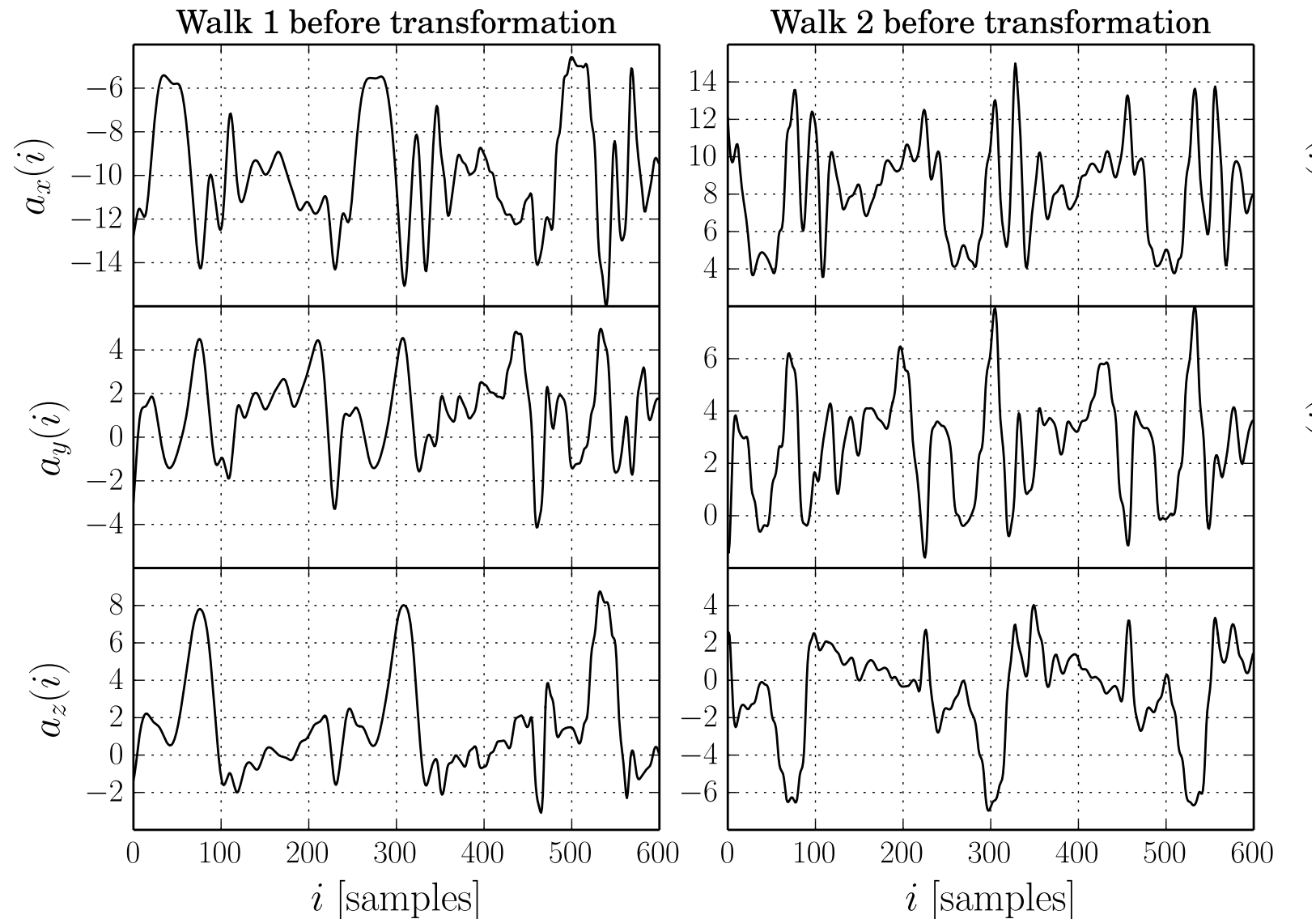


Reference system transform

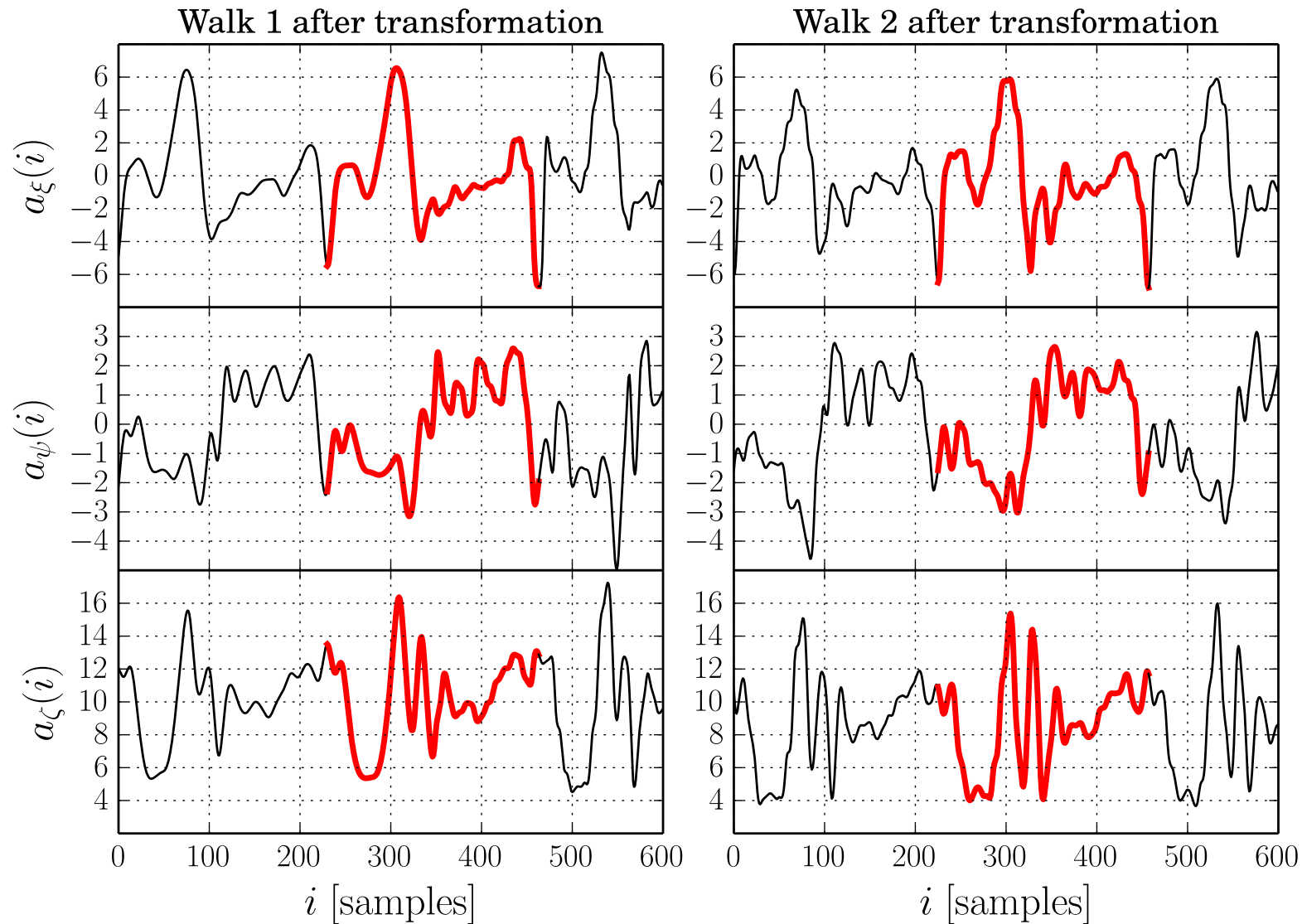


 new reference system versors

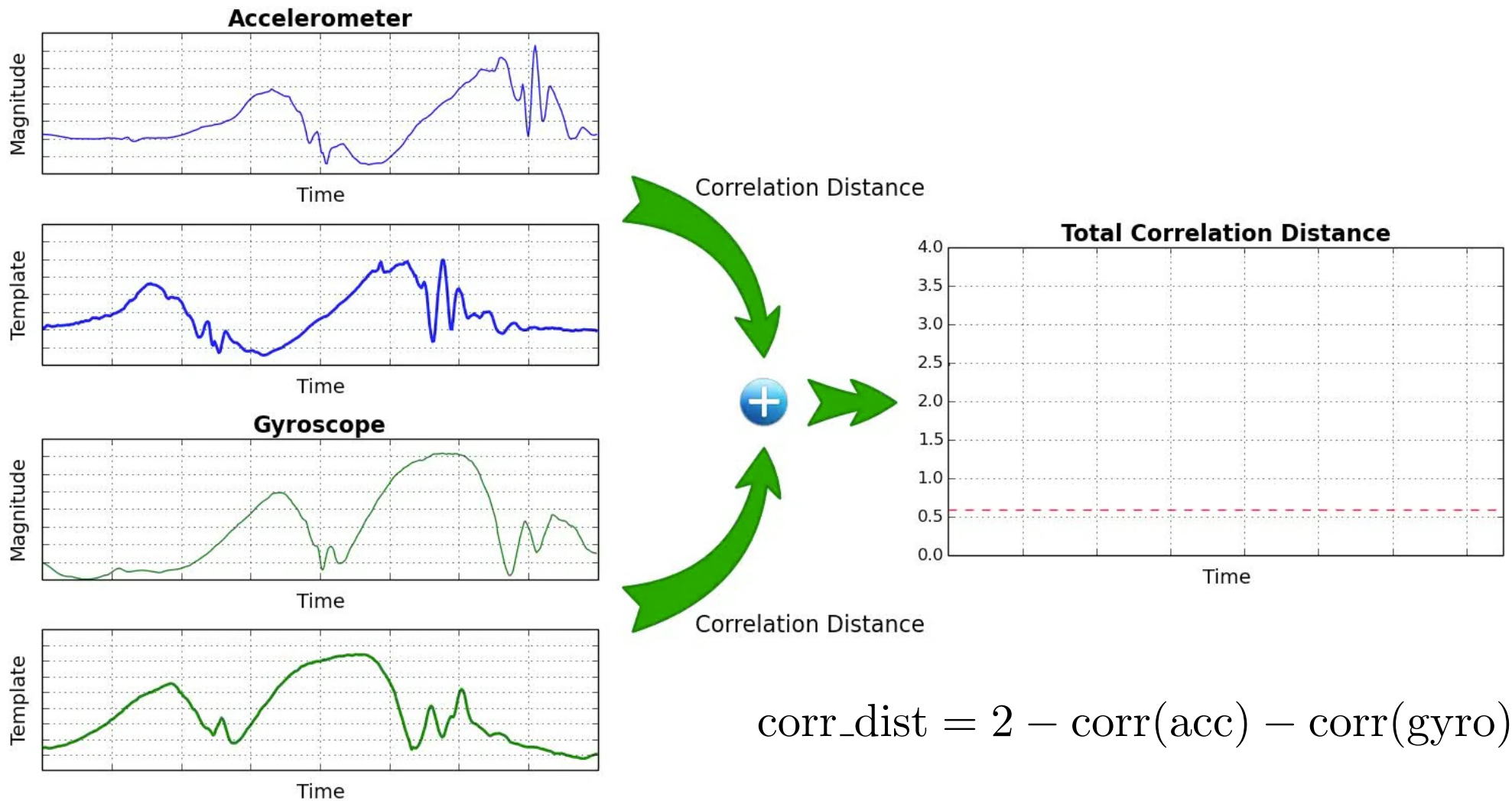
Walking pattern before transform



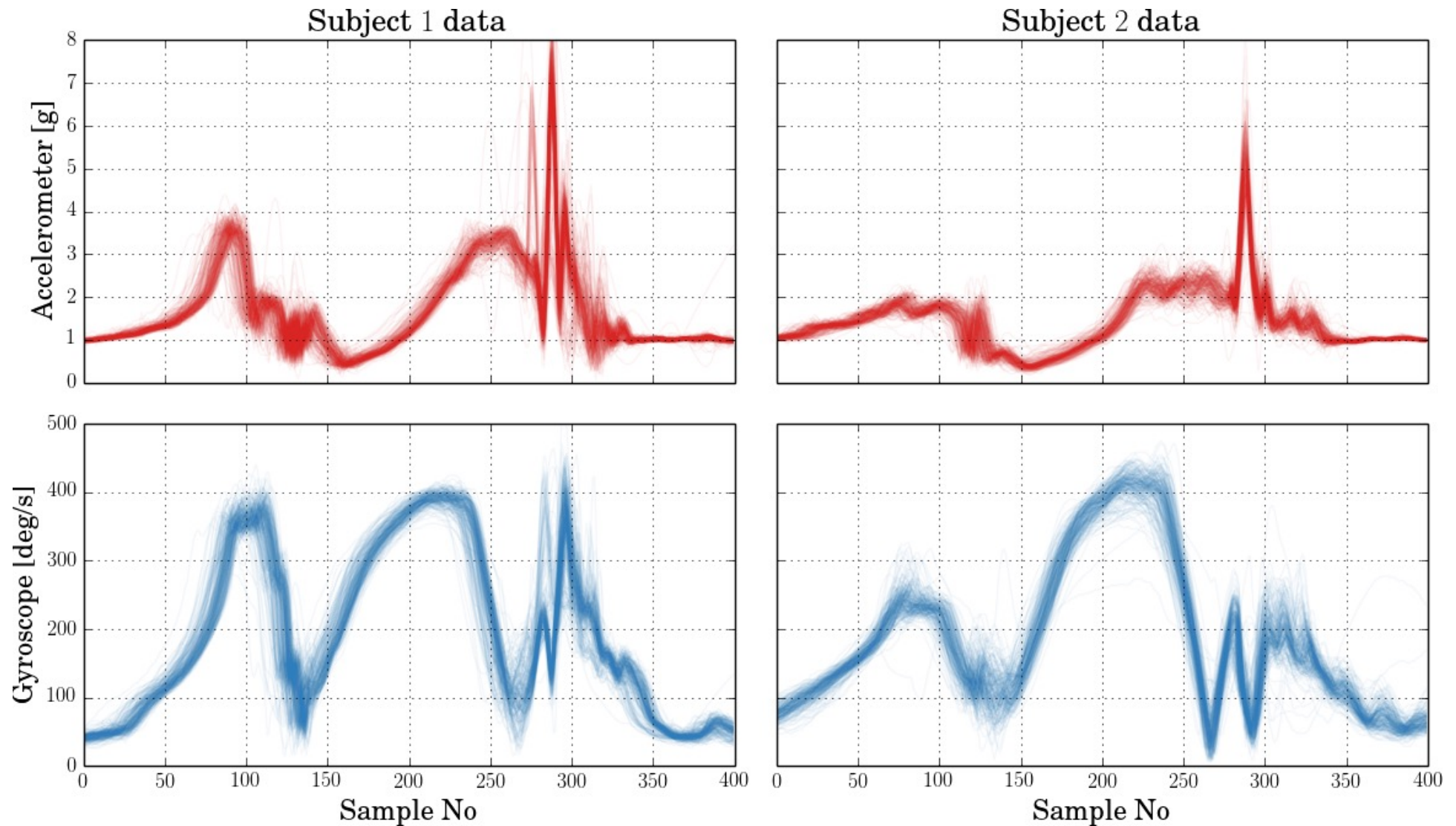
Walking pattern after tranform



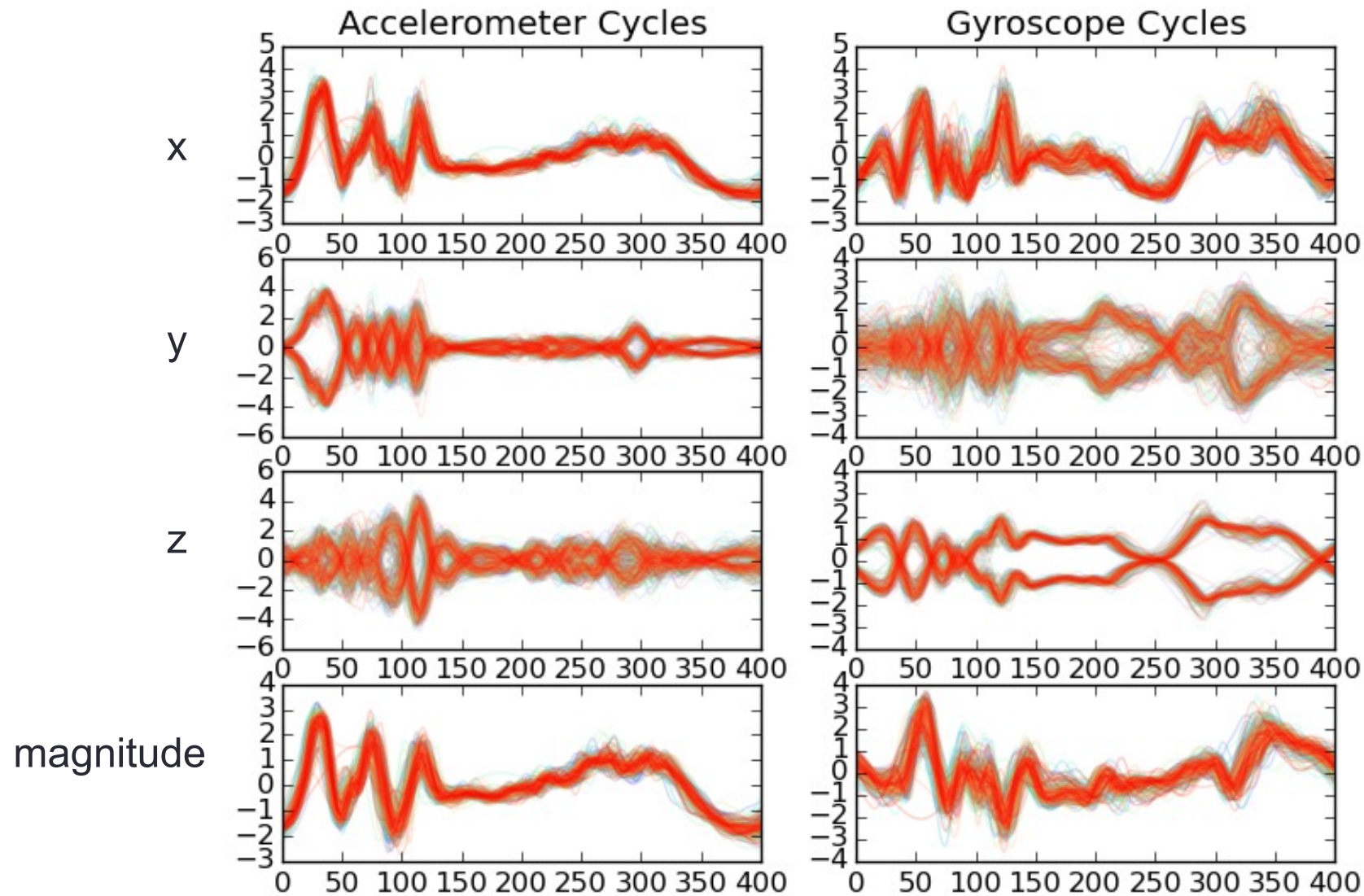
Template-based segmentation



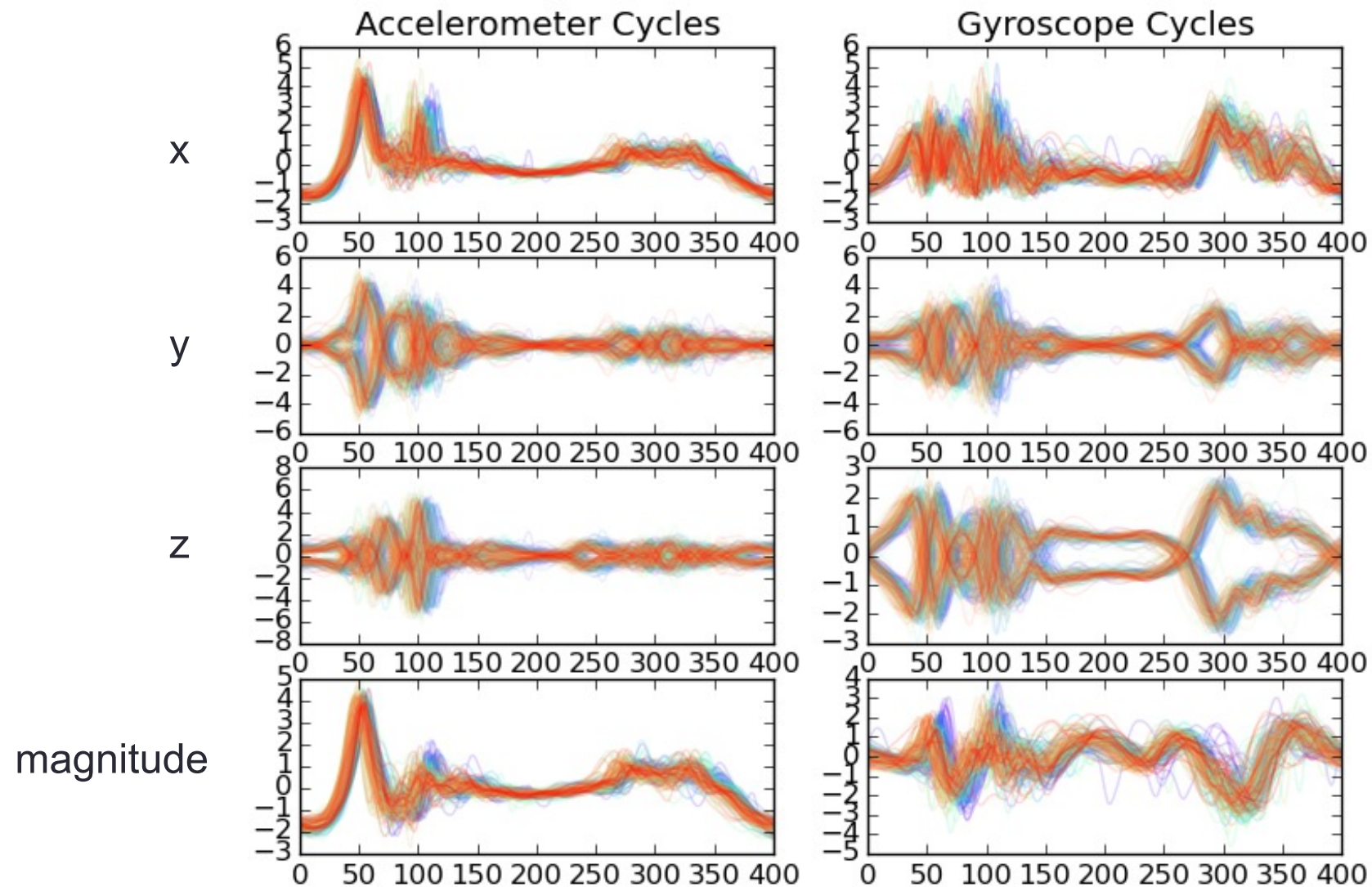
Walking pattern examples: two subjects



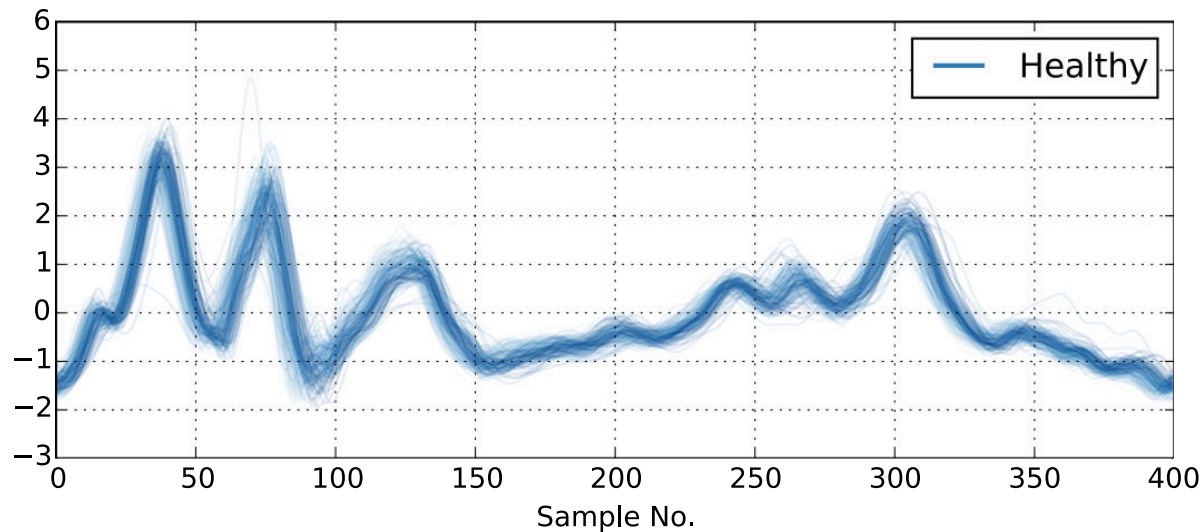
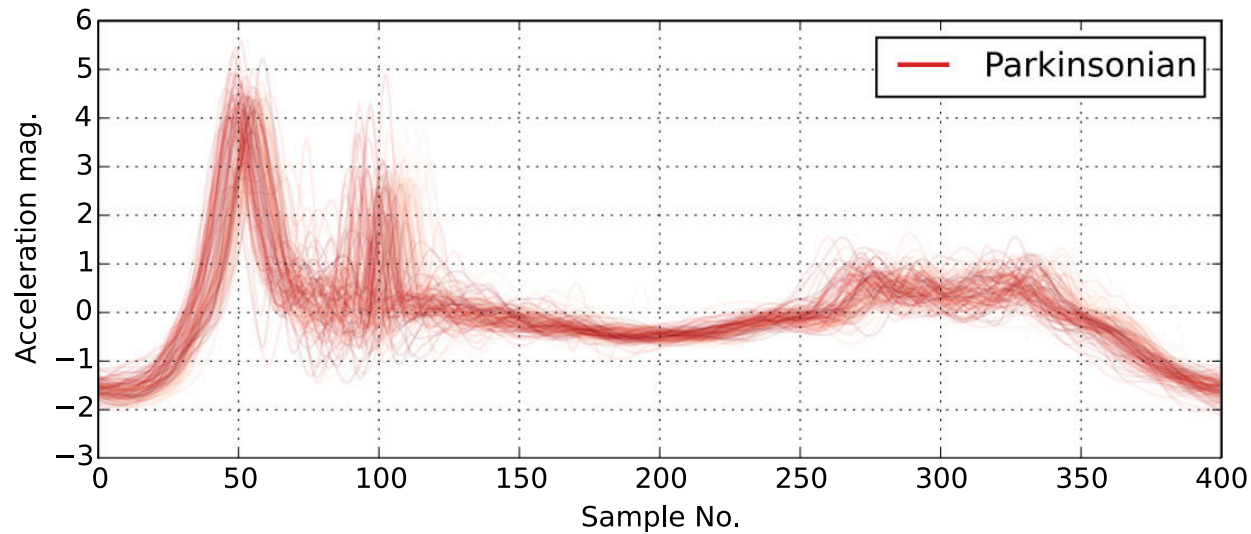
Healthy young male



Parkinsonian

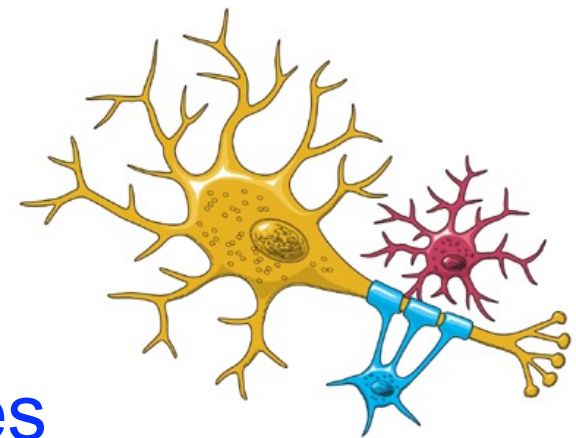


Healthy vs Parkinsonian



Convolutional Neural Network (CNN)

- **Supervised training (50 subjects)**
 - 10 minutes of walk, at least 2 sessions each
 - 200 walking cycles/subject used for training (correspond to approx. 2 minutes of walking time)
- **Smartphones**
 - Asus Zenfone 2,
 - Samsung S3 Neo,
 - Samsung S4,
 - LG G2, LG G4,
 - Google Nexus 5
- **CNN to extract of relevant features**
 - Automatic feature engineering



CNN – conv layer 1

- **CL1** We do not capture any correlation among different accelerometer and gyroscope axes
- 20, 1D kernels of size 1 x 10 (stride=5): processing each input vector separately. For each walking cycle, input vectors are (normalized to a fixed length):

$$a_x, a_y, a_z, |a|, \text{gyr}_x, \text{gyro}_y, \text{gyro}_z, |\text{gyro}|$$

- Activation functions are linear

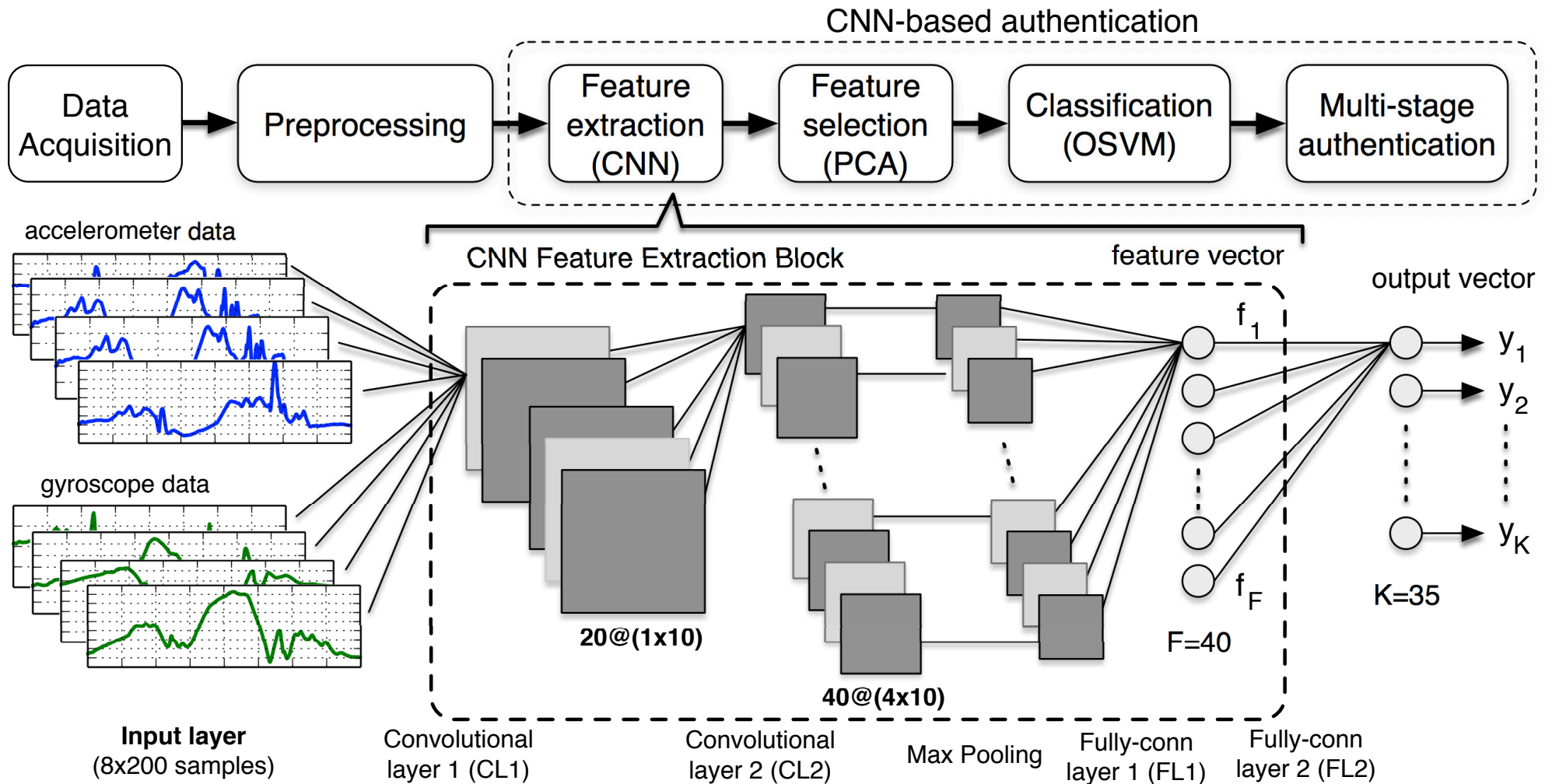
CNN – conv layer 2

- **CL2** With the second convolutional layer we seek discriminant and class-invariant features.
- 50, 2D kernels of size 3 x 10 (vertical stride=1, horizontal stride=3): the cross-correlation among input vectors is considered
- **tanh**: non-linear activation functions
- **Max pooling**: applied to
 - reduce the dimensionality, increase spatial invariance
 - output size is halved, takes the maximum between every 2 output values from previous CONV

Fully connected layers 2 and 3

- **FL2** fully connected layer with **tanh** activations
 - has **F** output neurons
 - These are the **features extracted by the network**
- **FL3** fully connected layer with **softmax** activations
 - has **K** output neurons (**K** is the number of classes)
 - one for each user in the **training dataset**

CNN architecture



One-Class SVM [Schölkopf-01]

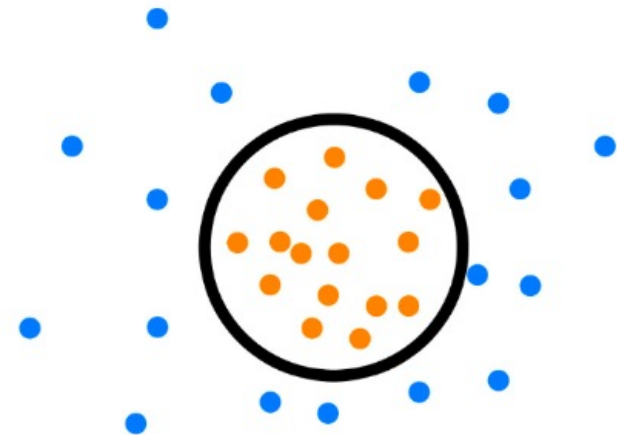
- Only target class (orange) data is available
- Training finds the *boundary* (tick circle)
- Classification output: **score**
 - Distance from the boundary

- **Training**

- We only used data from target user

- **Test**

- Data from the negative class are also used
 - Performance assessment

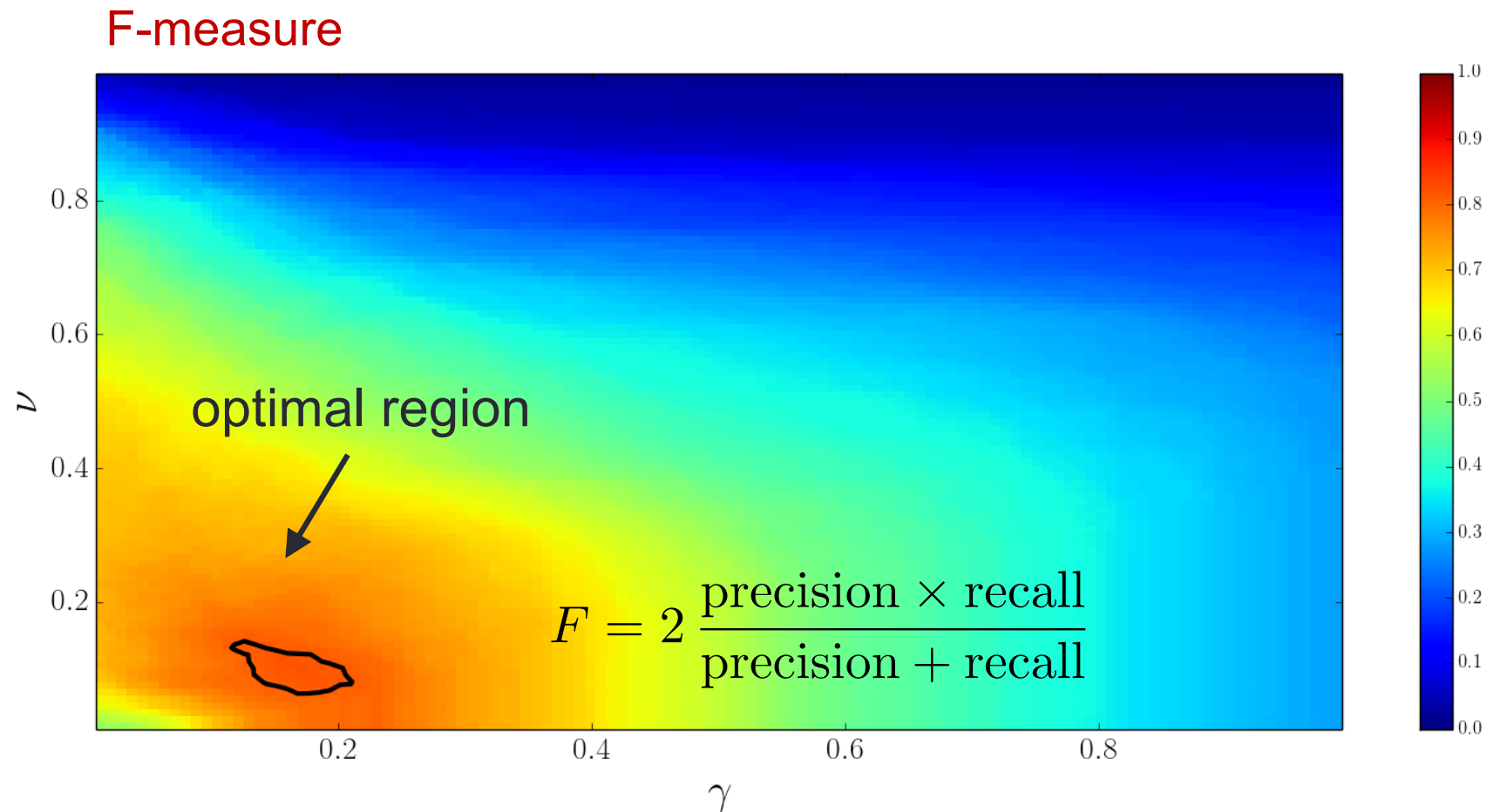


[Schölkopf-01] B. Schölkopf et al., “Estimating the Support of a High-Dimensional Distribution,” *Neural Computation Journal*, Vol. 13, No. 7, July 2001.

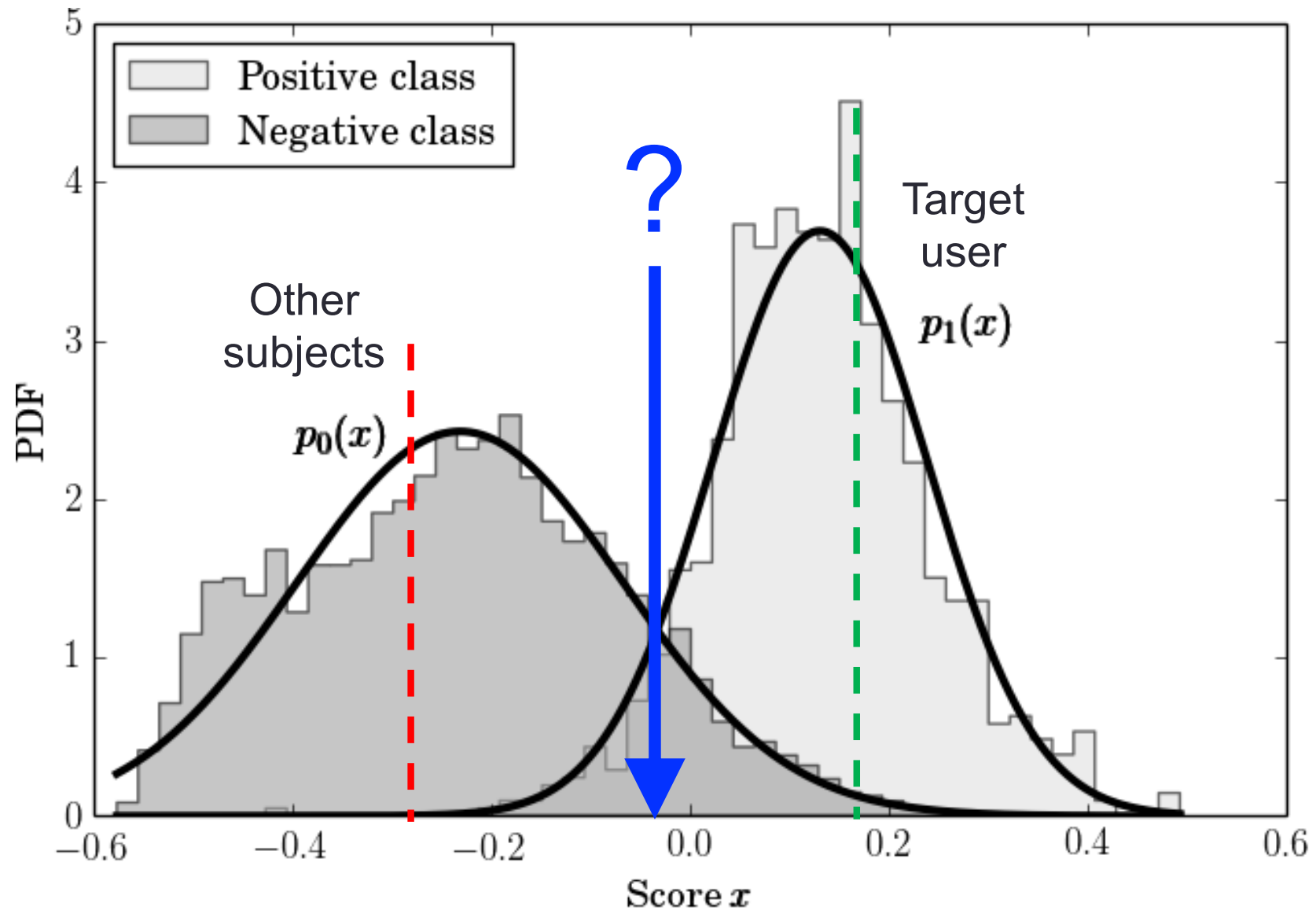
OSVM –parameters setting

Precision: no. of true positives classified as positives / tot. classified as positives

Recall: no. of true positives classified as positives / tot. no. of true positives



One-Class SVM (test output)



“I got the light bulb”



- **A single walking cycle**
 - Single pass through the network
 - Single SVM score
 - Unclear answer if happens to be in between the two pdfs
- What about observing multiple subsequent scores
 - Will they all fall in the problematic region (where the pdfs intersect)?
 - Probably not 😊
- Accumulating scores from multiple cycles
 - Should increase our likelihood of detecting the user

“I got the light bulb”



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- What about observing multiple subsequent scores
 - Will they all fall in the problematic region (where the pdfs intersect)?
 - Probably not 😊
- Accumulating scores from multiple cycles
 - Should increase our likelihood of detecting the user
 - But, how can we do this?

Abraham Wald

- American mathematician
- Born in Cluj, Romania
- Contributed in
 - decision theory,
 - geometry,
 - econometrics
- Worked @ Columbia University
- Founded the field of **statistical sequential analysis**

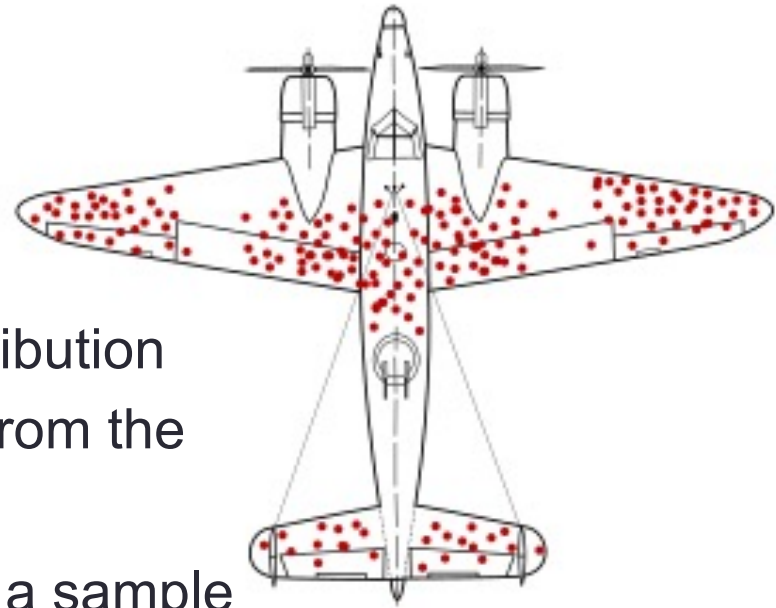


October 31, 1902 –
December 13, 1950

[W47] Abraham Wald, “Sequential analysis,” Dover, New York, NY, US, 1947.

Abraham Wald

- During **world war II**
 - Wald lead a group to study the distribution of damages from planes returning from the battlefield
 - He realized that such distribution is a sample **conditioned on the survival of the airplanes** (Bayes again...)
 - Taking care of this **bias** revealed that
 - The airplanes that were lost in battle were mainly hit in the undamaged spots/areas of those that returned
 - So the recommendation was to add special protection to the undamaged areas of the returning planes
 - This is an example of **survivorship bias**
 - As crucial data from fatally hit planes is ignored



Sequential decision making (1/2)

Sequential Probability Ratio Test (SPRT) [W47]

$$p_0(x) = p(x|H_0)$$

H_0  OTHER USERS' CLASS

$$p_1(x) = p(x|H_1)$$

H_1  TARGET CLASS

- Samples acquired sequentially, one at a time
 - are of either class H_0 (pdf $p_0(x)$) or H_1 (pdf $p_1(x)$)
 - are **i.i.d.**
- For the extension to correlated samples, see [TNB15]

[W47] A. Wald, “Sequential analysis,” Dover, New York, US, 1947.

[TNB15] A. Tartakovsky, I. Nikiforov, M. Basseville, “Sequential Analysis Hypothesis Testing and Changepoint Detection,” CRC Press, 2015.

Sequential decision making (2/2)

Sequential Probability Ratio Test (SPRT) [W47]

$$H_1 \rightarrow p_1(x) = p(x|H_1)$$

TARGET CLASS

$$H_0 \rightarrow p_0(x) = p(x|H_0)$$

OTHER CLASS

LOG-LIKELIHOOD ratio:

$$S_n = \log \left(\frac{p(x_1, \dots, x_n | H_1)}{p(x_1, \dots, x_n | H_0)} \right) = \sum_{i=1}^n \log \left(\frac{p_1(x_i)}{p_0(x_i)} \right)$$

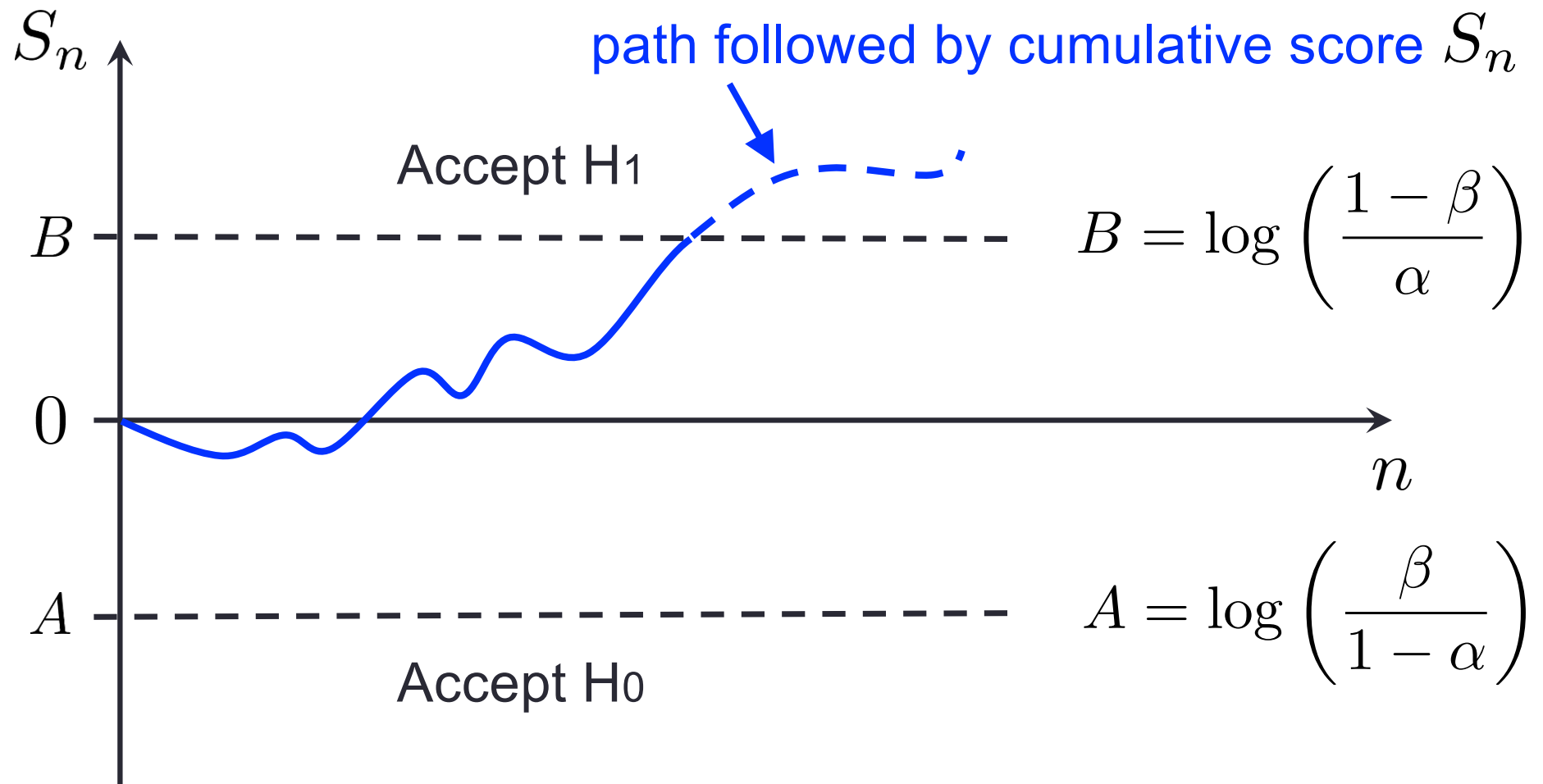
β = probability of accepting H_0
when H_1 is true

α = probability of accepting H_1
when H_0 is true

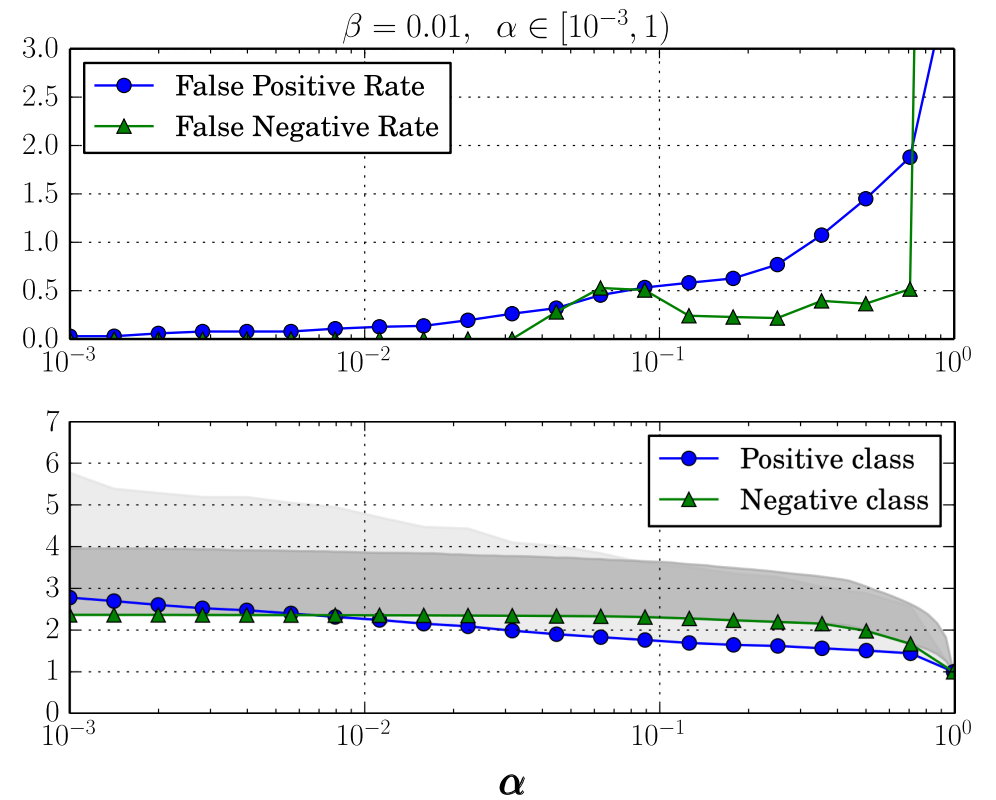
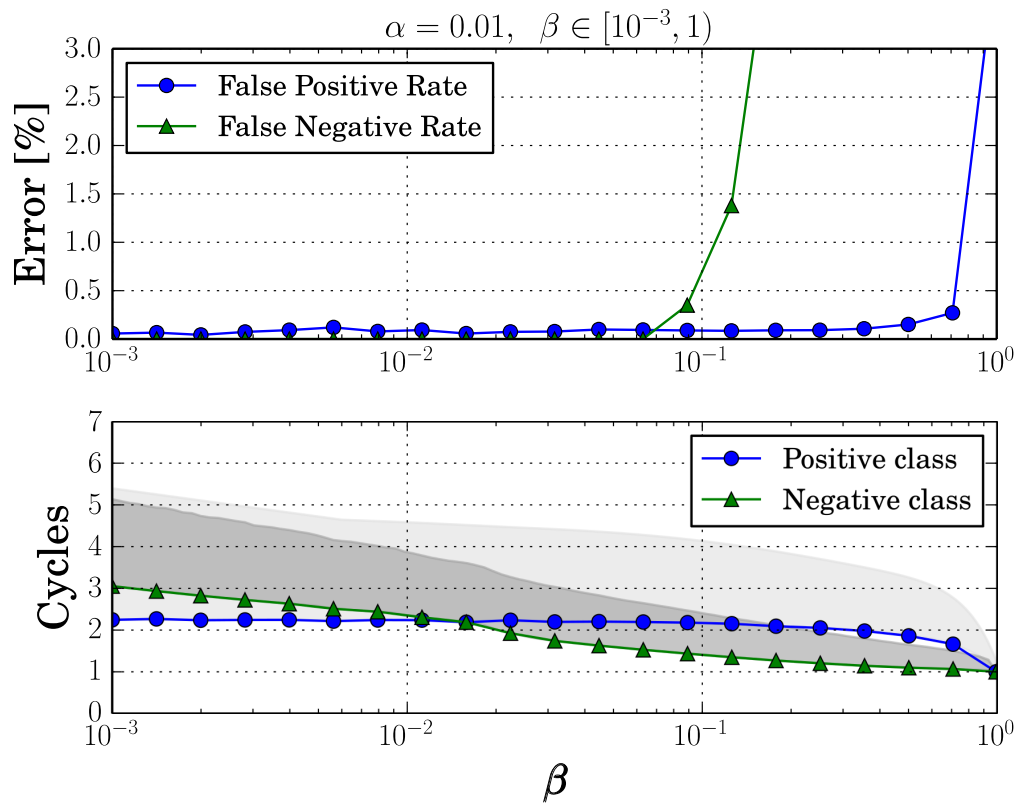
[W47] A. Wald, "Sequential analysis," Dover, New York, NY, US, 1947.

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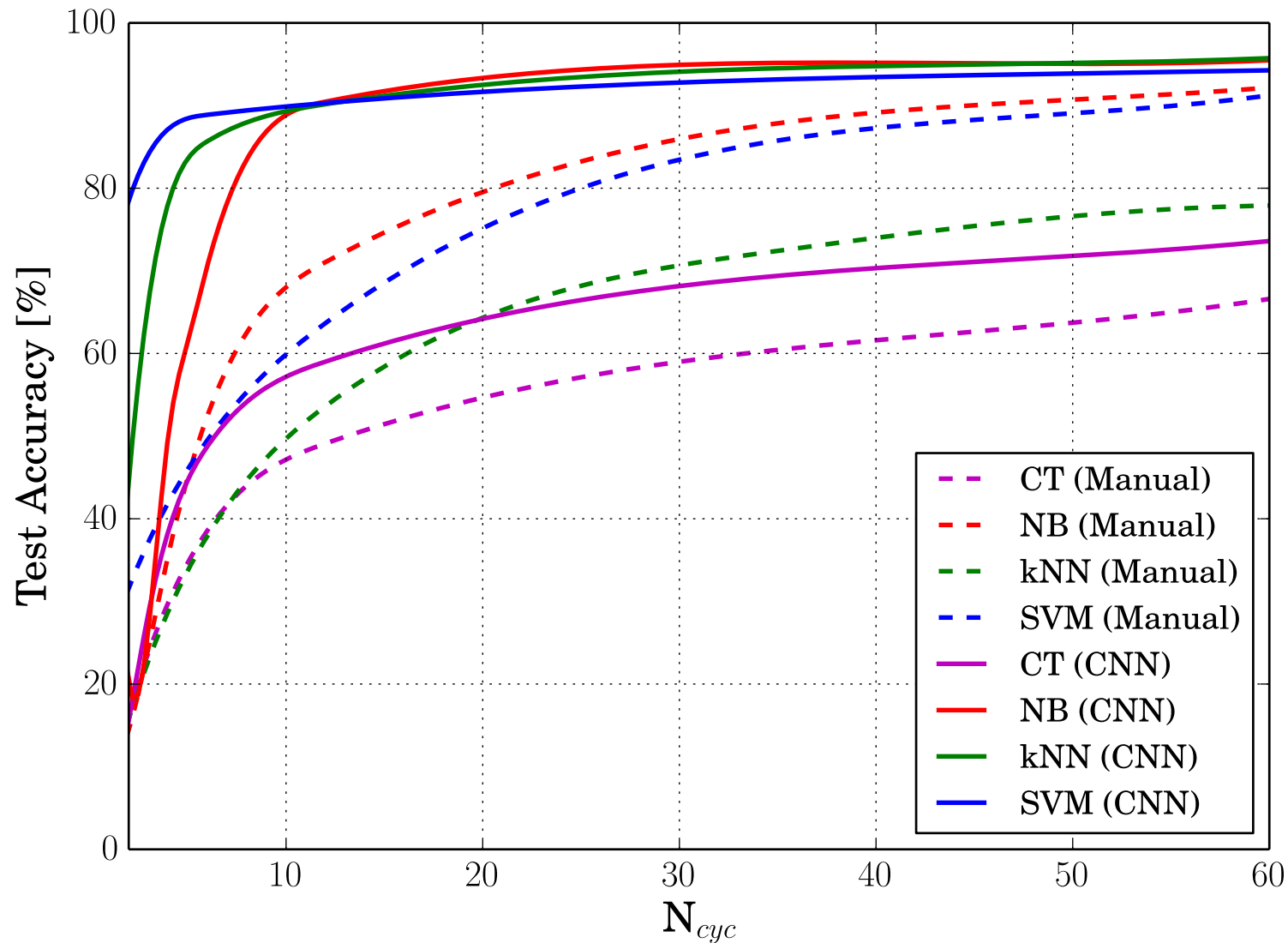
Sequential analysis



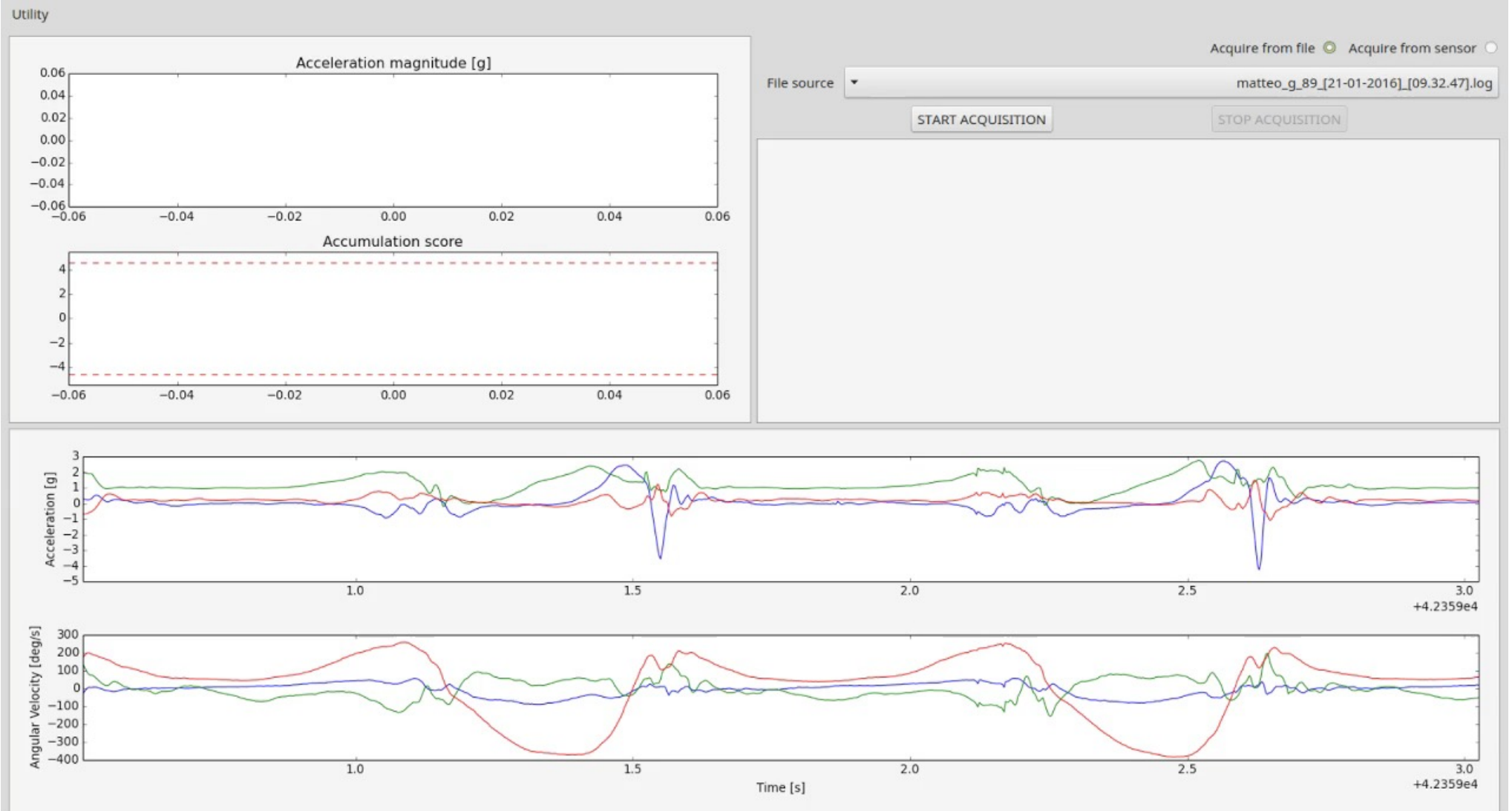
Final results



Feature extraction CNN vs SOTA



Demo



Bibliography

[Schölkopf-01] B. Schölkopf et al., “Estimating the Support of a High-Dimensional Distribution,” *Neural Computation Journal*, Vol. 13, No. 7, July 2001.

[W47] A. Wald, “Sequential analysis,” Dover, New York, NY, US, 1947.

[TNB15] A. Tartakovsky, I. Nikiforov, M. Basseville, “Sequential Analysis Hypothesis Testing and Changepoint Detection,” CRC Press, 2015.

[GR18] M. Gadaleta, M. Rossi, “IDNet: Smartphone-based Gait Recognition with Convolutional Neural Networks,” *Pattern Recognition*, Vol. 74, Pages 25–37, February 2018.

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